CSC1024 Programming Principles
Coding Exercise 1

1. Write a program that prompts the user to enter five test scores and then prints the average of the test scores.

2. Write a program that converts RM to SGD (Singapore Dollar). On 1st May 2020, the exchange rate is such that it requires RM 3.04 to obtain 1 SGD. Prompt the user to enter an RM amount and print the amount in SGD.

3. Light travels at 3 $*10^8$ meters per second. Write a program that calculates and displays the value of a light-year.
   A light-year: distance a light beam travels in one year.

4. Write a program that prompts the user to enter the length and height of a rectangle, and prints the area and perimeter of the rectangle.

5. Write a program that accepts an object's mass (in kilograms) and velocity (in meters per second) as inputs and then outputs its momentum and kinetic energy given the following information
   - The momentum is its mass multiplied by its velocity.
   - The kinetic energy is: $(1/2)mv^2$.

6. Write a program that requests the radius of a sphere (a floating-point number) as input from the user and then outputs the sphere's diameter, circumference, surface area, and volume.

7. Write a program that splits a restaurant bill among a group of friends. The program prompts the user to input the total of the bill and the number of friends splitting the bill. The program first calculates and adds a 10% service charge to the total, and then calculates and adds a 6% GST to the total (including the service charge). The program then outputs the amount to be paid by each friend.

8. Write a program that takes as inputs the hourly wage, total regular hours, and total overtime hours and displays an employee's total weekly pay given that:
   a. An employee's total weekly pay equals the hourly wage multiplied by the total number of regular hours plus any overtime pay.
   b. Overtime pay equals the total overtime hours multiplied by 1.5 times the hourly wage.